Noether Sheet 6

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April 11, 2026

These were my solutions to the Noether Mentoring Scheme Sheet 6. Problem statements are abridged.

Problem 1

The fibonacci sequence is the sequence $1, 1, 2, 3, 5, 8, 13, 21, \ldots$, where each term after the first two terms is the sum of the previous two.

(a) Show that there are infinitely many Fibonacci numbers that are the mean of two different Fibonacci numbers.

(b) Show that there are infinitely many Fibonacci numbers that are the mean of three different Fibonacci numbers.

Solution

(a) Let F_n denote the n-th Fibonacci number. Call a Fibonacci number *special* if it can be expressed as the mean of two different Fibonacci numbers. After some exploration one may find that $F_5 = \frac{1}{2}(F_3 + F_6)$ and similarly, $F_{11} = \frac{1}{2}(F_9 + F_{12})$. These results motivate the following claim,

Claim 1: For positive integer n,

$$F_{3n+2} = \frac{1}{2}(F_{3n} + F_{3n+3}).$$

Proof: The proof is as follows,

$$\begin{aligned} F_{3n} + F_{3n+3} &= 2F_{3n+2} \\ F_{3n} + (F_{3n+1} + F_{3n+2}) &= F_{3n+2} + F_{3n+2} \\ F_{3n} + F_{3n+1} &= F_{3n+2} \end{aligned}$$

which is true by definition of the Fibonacci numbers. □

Thus since there are infinitely many $n \in \mathbb{N}$, using claim 1 we can show that infinitely many Fibonacci numbers are special. □

(b) Call a Fibonacci number *quirky* if it can be expressed as the mean of three different Fibonacci numbers. Simimlarly to (a) after a bit of exploration we may find that $F_6 = \frac{1}{3}(F_4 + F_6 + F_7)$ and $F_{14} = \frac{1}{3}(F_{12} + F_{14} + F_{15})$.

Claim 2: For positive integer n,

$$F_{8n-2} = \frac{1}{3}(F_{8n-4} + F_{8n-2} + F_{8n-1}).$$

Proof: The proof is as follows,

$$\begin{aligned} F_{8n-4} + F_{8n-2} + F_{8n-1} &= 3F_{8n-2} \\ F_{8n-4} + F_{8n-1} &= 2F_{8n-2} \\ F_{8n-4} + (F_{8n-2} + F_{8n-3}) &= 2F_{8n-2} \\ F_{8n-4} + F_{8n-3} &= F_{8n-2} \end{aligned}$$

which is true due to the definition of the Fibonacci numbers. □

Thus since there are infinitely many $n \in \mathbb{N}$, using claim 2 we can show that infinitely many Fibonacci numbers are quirky. ■

Problem 2

Solve the simultaneous equations

$$
\begin{aligned}
x + \sqrt{y} &= 13, \\
xy &= 90.
\end{aligned}
$$

Solution

Square the first equation to find that $y = (13 - x)^2$. Sub into the second equation to get

$$
\begin{aligned}
x(13 - x)^2 &= 90 \\
x^3 - 26x^2 + 169x - 90 &= 0 \\
(x - 10)(x^2 - 16x + 9) &= 0
\end{aligned}
$$

which solves to give $x = 10, 8 + \sqrt{55}, 8 - \sqrt{55}$. But now one can simply check that the only solutions are $(x, y) = (10, 9), (8 - \sqrt{55}, 80 + 10\sqrt{55})$. ■

Problem 3

Let ABC be a triangle.

(a) Prove that the perpendicular bisectors of the three sides meet at a point. This is the *circumcentre* of $\triangle ABC$, usually denoted O.

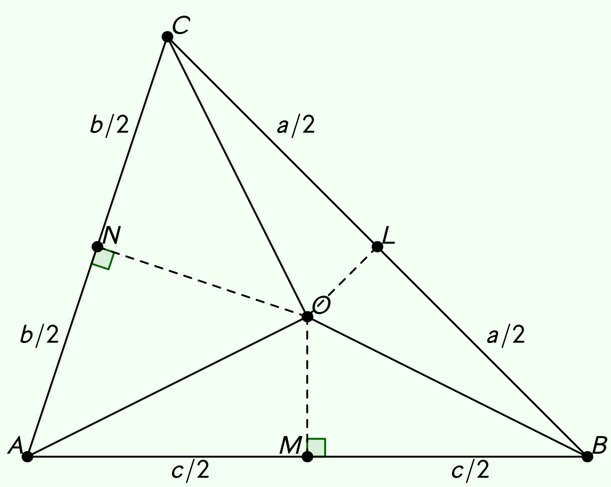
Prove also that the circle through A, B and C has it's centre at O.

(b) Prove that the angle bisectors of $\angle A, \angle B$ and $\angle C$ meet at a point. This is the *incentre* of $\triangle ABC$, usually denoted I.

(c) Prove that the altitudes of the triangle meet at a point. This is the *orthocentre* of the $\triangle ABC$, usually denoted H.

Solution

(a) Construct points L, M, N to be the midpoints of the sides BC, AB, AC respectively. Let the side lengths be a, b, c is the usual convention. Let O be the intersection of the perpendicular bisectors of AC, AB. Such a point must exist because two lines intersect if they aren't parallel, since AC and AB aren't parallel, their perpendicular bisectors aren't parallel either. We have the diagram,



And our goal is to show that $\angle OLB = 90°$.

Claim 1: If $OC = OB$ then this implies $\angle OLB = 90°$.

Proof: Let $\angle OLB = \theta$. Due to cosine rule and angles on a straight line summing to 180,

$$OC^2 = \left(\frac{a}{2}\right)^2 + OL^2 - 2\left(\frac{a}{2}\right)(OL)\cos(180 - \theta)$$
$$OB^2 = \left(\frac{a}{2}\right)^2 + OL^2 - 2\left(\frac{a}{2}\right)(OL)\cos(\theta)$$

and so if $OB = OC$ then $\cos(\theta) = \cos(180 - \theta) = -\cos(\theta)$. But this means that $\cos\theta = 0$ and since $\theta \in (0, 180)$ we have $\theta = 90°$ only. $\square$

Denote $x = OB, y = OC, h = OM$. By Pythagoras,

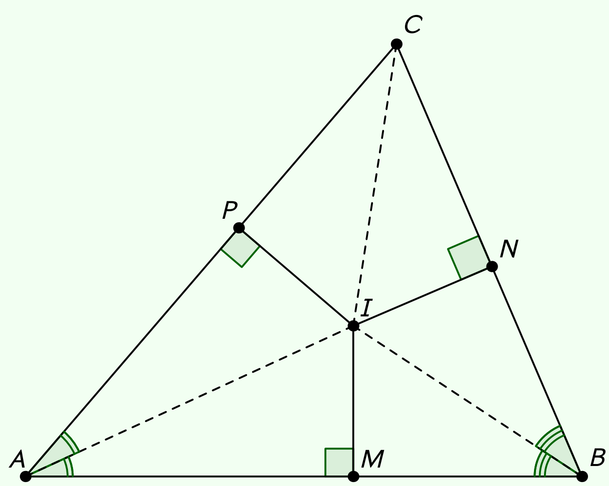
$$\begin{aligned} x^2 &= h^2 + \left(\frac{c}{2}\right)^2 \\ y^2 &= \left(\frac{b}{2}\right)^2 + ON^2 \\ &= \left(\frac{b}{2}\right)^2 + \left[OA^2 - \left(\frac{b}{2}\right)^2\right] \\ &= OA^2 = \left(\frac{c}{2}\right)^2 + h^2 \end{aligned}$$

So this must mean that $x = y$ and due to claim 1 we are done. Note that $OA^2 = (c/2)^2 + h^2$ also by Pythagoras and so $OA = OB = OC$. Since the point O is equidistant from all three points, the circle through A, B, C has centre O and radius OA. □

(b) Let I intersection of the angle bisectors B_A of $\angle A$ and B_B of $\angle B$. We need to show that such a point must exist.

WLOG assume the point configuration as seen in the diagram. If two lines aren't parallel then they must intersect at a point. Suppose for the sake of contradiction that those two bisectors are parallel. Let θ be the angle B_A makes with the side AB (measuring anti-clockwise), since B_B is parallel it also makes an angle of θ of with the side AB (measuring anti-clockwise). That means that measuring clockwise the angle it makes is $180 - \theta$. But this angle is exactly $\frac{1}{2}\angle B$ so $\angle B = 360 - 2\theta$, similarly $\angle A = 2\theta$. But clearly this means that the interior angle sum in $\triangle ABC$ is greater than 180, a contradiction.

Let P, N, M be the feet of the perpendiculars from I to sides AC, BC, AB. Our goal is to show that $\angle PCI = \angle NCI$.



Claim 1: A point A lies on the bisector of PQ and PR if and only if the perpendiculars from A to PQ, PR are the same length.

Proof: Let (1) be the statement "A point A lies on the bisector of PQ and PR". Let (2) be the statement "The perpendiculars from A to PQ, PR are the same length".

First we show that (1) $\implies$ (2). Let F_Q, F_R be the feet of the perpendiculars from A to PQ, PR respectively. Since $\angle APQ = \angle APR$ by assumption and $\angle PF_QA = \angle PF_RA = 90°$, and angles in a triangle sum to 180 means that their third angles are the same also, the triangles $\triangle APF_Q$ and $\triangle APF_R$ are similar by AAA criterion. In fact since both triangles share the hypotenuse AP, the two triangles are congruent. This gives $AF_Q = AF_R$.

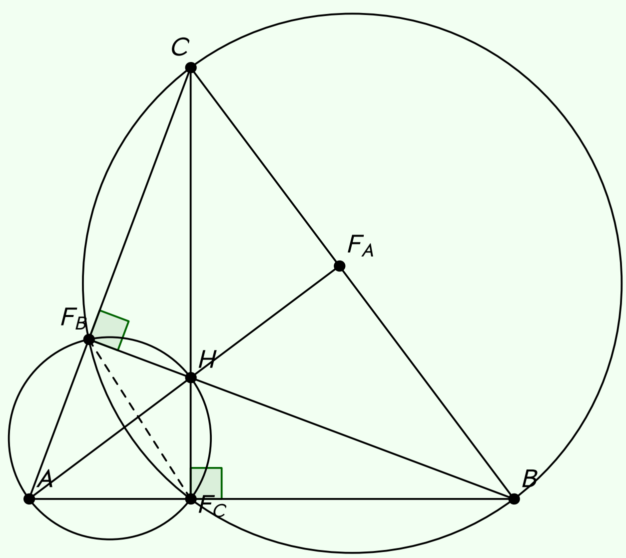
Next we show that (2) $\implies$ (1). Define F_Q, F_R as we did before. Since $\angle AF_QP = \angle AF_RP = 90°$, $AF_R = AF_Q$ by assumption and the triangles $\triangle AF_QP$ and $\triangle AF_RP$ both share hypotenuse AP, the two triangles are congruent by SAS criterion. But this means that $\angle APF_R = \angle APF_Q$. □

Now using the fact that two equal angles in two triangles implies that their third angles are equal, the triangles $\triangle PIA$ and $\triangle MIA$ are similar and so are the triangles $\triangle MIB$ and $\triangle NIB$. Moreover since each pair of triangles shares a hypotenuse the each pair of two triangles are actually congruent. This allows us to conclude $IM = PI = NI$.

Now we have $NI = PI$ so claim 2 tells us that I lies on the angle bisector of (PC, CN). That means that CI bisects $\angle PCN$ and so $\angle PCI = \angle NCI$ as desired. □

(c) Let H be the intersection of the altitudes through C and through B. Two lines intersect if they aren't parallel. The altitude through C is perpendicular to AB, and the altitude through B is perpendicular AC, so if the two altitudes were parallel then this would imply that $AC \parallel AB$ which is impossible for a non-degenerate triangle.

Let F_B, F_C be the feet of the altitudes through B, C respectively. Let F_A be the intersection of the line through (A, H) and side BC. Our goal is to show $\angle AF_AB = 90°$.



Claim 1: AF_BHF_C is a cyclic quadilateral.

Proof: We have $\angle AF_CH = \angle AF_BH = 90°$, so these two opposite angles sum to 180. Now since angles in a quadrilateral sum to 360 this forces $\angle F_BAF_C + \angle F_BHF_C = 180°$, which means that AF_BHF_C is cyclic. □

Claim 2: CF_BF_CB is a cylic quadrilateral.

Proof: Firstly notice that $\angle ACF_C = 90 - \angle A$. Then since AF_BHF_C is cyclic, we know that $\angle F_BHF_C = 180 - \angle A$

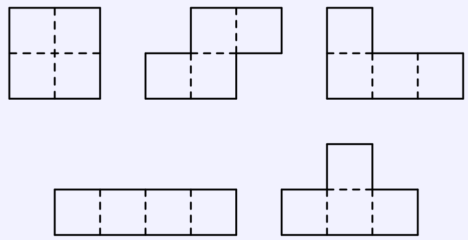
which means that $\angle F_C HB = \angle A$ due to angles on a straight line sum to 180. Then we get that $\angle HBF_C = 90 - \angle A$.

Now the key step is to use the fact that $\angle F_B BF_C = \angle F_B CF_C$, and then by the converse of "Angles subtending the same segment are equal" it must be the case that F_B, F_C, B, C all lie on the same circle. □

Now since $CF_B F_C B$ is cyclic, we know that $\angle F_B F_C B = 180 - \angle C$. Since $\angle HF_C B$ is right-angled we deduce $\angle F_B F_C H = 90 - \angle C$. Now consider the cyclic quadrilateral $AF_B HF_C$. Since angles subtending the same segment are equal we have $\angle F_B AH = 90 - \angle C$. Finally we can look at $\triangle CF_A A$ and deduce that $\angle CF_A A = 180 - (\angle C + 90 - \angle C) = 90°$. as desired. ■

Problem 4

The five tetrominoes are shown below. They can be turned either way up.

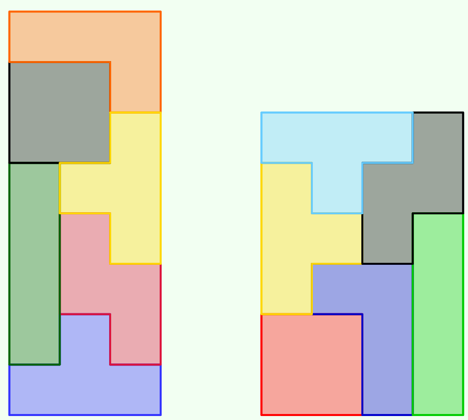


(a) Show that a 4×6 and a 3×8 rectangle can be tiled by six tetrominoes, five of them different. Which tetromino have you used twice?

(b) Show that no rectangle can be tiled by the set of five each used once.

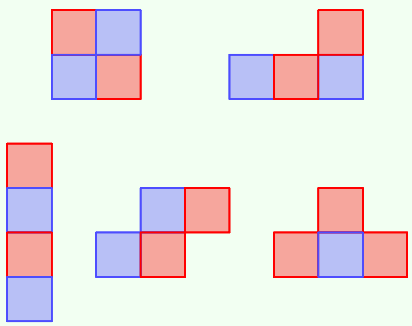
Solution

(a) Here are the two rectangles.



In each case we used the "T" tetromino twice. □

(b) Observe the following diagram.



Thus if we were to tile the tetrominoes on a chessboard, then every tetromino except the "T" tetromino will cover 2 white squares and 2 black squares. The "T" tetromino covers 3 white squares and 1 black square or 3 black squares and one white square. In any case if we tile all five tetrominoes together on a rectangular chessboard we will cover 2 more white squares than black squares (or 2 more black squares than white squares). However this is impossible, since a chessboard has an equal number of white and black squares. Therefore it is not possible to tile a rectangle using each tetromino exactly once. ■

Problem 5

(a) Suppose the numbers $1, 2, \ldots, n$ are to be split into three non-empty subsets. How many ways are there to achieve this?

(b) What happens if we insist that each subset consists of a run of at least one consecutive number?

Solution

(a) Clearly $n \geq 3$. Suppose the first subset has size a the second has size b and the third has size c.

- Since $b, c \geq 1$ we find that a is at most $n-2$.
- Since $c \geq 1$ we have that $a + b \leq n - 1$ and so b is at most $n - 1 - a$.
- Finally there is exactly one choice for c that is $c = n - a - b$.

Putting this together, the number of choices turns out to be

$$S = \sum_{a=1}^{n-2}\left[\binom{n}{a} \times \sum_{b=1}^{n-a-1}\binom{n-a}{b} \times 1\right].$$

By Binomial theorem,

$$\sum_{b=0}^{n-a}\binom{n-a}{b} = (1+1)^{n-a} = 2^{n-a}$$

and so we can re-write S as

$$S = \sum_{a=1}^{n-2}\left[\binom{n}{a}(2^{n-a}-2)\right].$$

We apply a similar observation

$$\sum_{a=0}^{n}\binom{n}{a}2^{-a} = \left(1+\frac{1}{2}\right)^n = \left(\frac{3}{2}\right)^n \quad \text{and} \quad \sum_{a=0}^{n}\binom{n}{a} = (1+1)^n = 2^n.$$

Thus

$$\begin{aligned} S &= 2^n\left[\left(\frac{3}{2}\right)^n - \binom{n}{0} - \binom{n}{n-1}\frac{1}{2^{n-1}} - \binom{n}{n}\frac{1}{2^n}\right] - 2\left[2^n - \binom{n}{0} - \binom{n}{n-1} - \binom{n}{n}\right] \\ &= [3^n - 2^n - 2n - 1] - 2\,[2^n - n - 2] \\ &= 3^n - 3(2^n) + 3 \end{aligned}$$

and so there are exactly $3^n - 3(2^n) + 3$ ways to split the numbers. □

(b) To split the numbers in this way we can think about placing two "dividers" in the spots underlined below

$$1 \_\ 2 \_\ 3 \_\ 4 \_\ 5 \_ \ldots \_\ (n-1) \_\ n$$

There are $(n-1)$ places to put the dividers so the answer is $\binom{n-1}{2}$. ■

Problem 6

(a) In a group of six academics, each pair corresponds only about one topic. Moreover, the only two topics are geometry and algebra. Prove that there must be three academics such that each pair of those three correspond with each other about the same topic.

(b) In a group of seventeen academics, each pair corresponds only about one topic. Moreover, the only three topics are geometry, algebra and number theory. Prove that there must be three academics such that each pair of those three correspond with each other about the same topic.

Solution

(a) *Claim 1: Consider the complete graph with 6 vertices, K_6. Colour every edge either red or blue. Then there must exist a monochromatic 3-cycle in K_6.*

Proof: We proceed by way of contradiction. On K_6 every vertex has degree 5. Consider vertex i. Suppose there are r red edges and b blue edges that connect to i. At least one of r, b must be ≥ 3 because otherwise

$$\deg i = 5 = r + b \leq 2 + 2 = 4$$

which is impossible. Suppose that the edges from vertex i to vertices a, b, c are all the same colour, say WLOG they are all red. Since there doesn't exist a monochromatic 3-cycle on K_6 all of the edges $a - b$, $b - c$ and $c - a$ must be blue. But now notice that there is a monochromatic 3-cycle between $a - b - c$ where all edges are blue. Contradiction. □

If we model each academic as a vertex on a graph, colour an edge red if a pair corresponds about geometry and colour an edge blue if a pair corresponds about algebra, then the problem reduces to showing that claim 1 is true, as we have done. □

(b) *Claim 2: Consider the complete graph with 17 vertices, K_{17}. Colour every edge either red, green or blue. Then there must exist a monochromatic 3-cycle in K_{17}*

Proof: We proceed by way of contradiction. On K_{17} every vertex has degree 16. Consider a vertex i. Suppose that there are r red edges, b blue edges and g green edges that connect to i. At least one of r, b, g must be ≥ 6 because otherwise

$$\deg i = 16 = r + b + g \leq 5 + 5 + 5 = 15$$

which is impossible. Suppose that the edges from vertex i to the vertices a, b, c, d, e, f are all the same colour, say WLOG they are red. We note that any edges between two of these vertices cannot be red, because there are no 3-cycles on K_{17}.

Thus the vertices a, b, c, d, e, f actually form a 2-coloured complete sub-graph with 6 vertices, since edge colours can only be blue or green. However due to claim 1, there must exist a monochromatic 3-cycle on this sub-graph, contradiction. □

If we model each academic as a vertex on a graph, colour an edge red if a pair corresponds about geometry and colour an edge blue if a pair corresponds about algebra and colour an edge green if a pair corresponds about number theory, then the problem reduces to showing that claim 2 is true, as we have done. ■

Comments: This is really the first time I've done any graph theory in maths, though previously I've come across graphs in computer science and through solving problems on atcoder.jp and codeforces.com etc. Actually today I was recommended a youtube video that was a lecture on Ramsey theory, and I ended up finding out that part (a) is equivalent to showing that $R(3,3) = 6$ and part (b) is equivalent to showing that $R(3,3,3) = 17$.

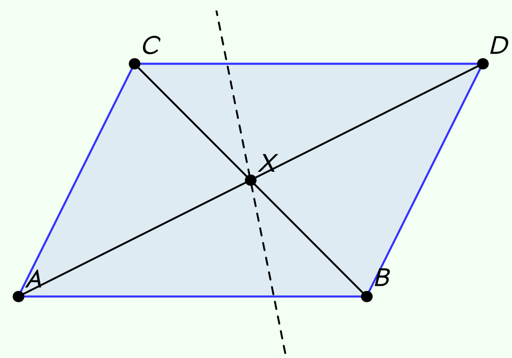
Problem 7

Let G be a convex quadrilateral. Show that there is a point X in the plane of G with the property that every straight line through X divides G into two regions of equal area if and only if G is a parallelogram.

Partial Solution

First we show the following statement to be true. If G is a parallelogram then there exists some point X, with the property that any line through X splits G into two regions of equal area. Let G have vertices $ABCD$.

Take $X = AD \cap BC$ like so.



Claim 1: BC and AD are lines of symmetry of G.

Proof: We show that $\triangle AXB \equiv \triangle CXD$.

- $AB = CD$ by properties of a parallelogram.
- $\angle BAX = \angle CDX$ because of alternate angles.
- $\angle ABX = \angle XCD$ because of alternate angles.

Thus the two triangles are indeed congruent by the ASA criterion. This allows us to conclude that $AX = DX$. Now since A, X, D are collinear, D is in fact the reflection of the point A across BC. Moreover points C, B are invariant under the transformation of reflection across BC, since they lie on this line. This means that G is in fact unchanged under this reflection and so BC is a line of symmetry of G. By a similar arguement we can show that AD is also a line of symmetry of G. □

Now consider some line l passing through X.

Case 1: The line l doesn't pass through any of A, B, C or D. We assume WLOG that l passes through sides AB and CD. We have already seen that $AX = DX$ and $CX = BX$. The points A, D lie on opposite sides of the line l, and the same is true for the points B, C. Now since A, X, D are collinear and l passes though X, D is the reflection of A across l. Similarly C is the reflection of B across l.

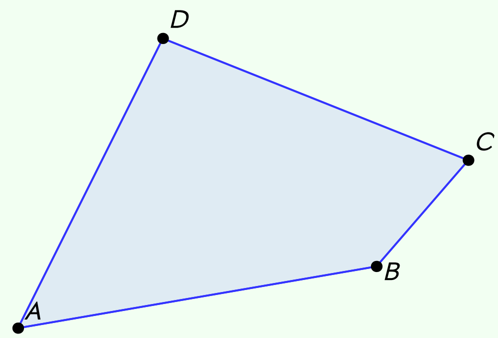
Suppose that l cuts through CD and AB at points H, K respectively. If we reflect $ACHK$ across l it maps onto $BDHK$. Since reflection transformations preserve areas, we have that the area of $ACHK$ and the area of $BDHK$ are equal.

Case 2: The line l passes through two diagonally opposite vertices. Then l either passes through BC or AD. Due to claim 1, we established that these are lines of symmetry of G, so the two regions formed map onto each other upon reflection across l. Again since reflections preserve area, the area of the two regions are equal.

Next we show the following statement to be true. For some convex quadrilateral G, if there exists a point X with the property that any line through X divides G into two regions of equal area then G must be a parallelogram. Call such a point X *special*.

We instead prove the contrapostive of this statement, which is logically equivalent. We show that if G is not a parallelogram, then there does not exist a special point X in the plane.

Let us suppose that G has vertices $ABCD$.



Claim 1: For any line l splitting G into two regions of non-zero area we can translate l onto precisely one line l' such that l' splits G into two regions of equal areas.

Proof: We may rotate the entire configuration so that l is parallel to the y-axis, because rotations leave lengths and areas unchanged. Now l splits G into a left region with area L and a right region with area R. Suppose the parallelogram has area 1. If we move l to be as rightmost as possible then $L \to 1$ and $R \to 0$, similarly if we move l to be as leftmost as possible then $L \to 0$ and $R \to 1$. Recalling that $L + R = 1$, by applying intermediate value theorem coupled with the fact that moving the line in one direction changes L, R monotonically there must be some unique point at which $L = R = 1/2$. $\square$

If I take some line M that splits G into two regions of equal area, then for any point P that doesn't lie on M, we can take the line through P that is parallel to M. Then since $P \neq M$ due to claim 1, this line doesn't split G into two regions of equal area. Thus any point P not lying on M isn't special.

We may make the exact same statement but this time replacing the line M with some line T that isn't parallel to M. So any point P not lying on T isn't special either. The only point remaining that could be special is the point $Y = T \cap M$, since it is the only point lying on both T and M.

The solution is nearly finished, we just need to patch up this line passing through Y case.

Problem 8

Let $p \equiv 3 \pmod 4$ be a prime.

(a) Suppose that $x^4 \equiv 1 \pmod p$. Prove that $x^2 \equiv 1 \pmod p$.

(b) Therefore show that there are no solutions to $x^2 + 1 \equiv 0 \pmod p$.

(c) Prove that there are an infinite number of primes $\equiv 1 \pmod 4$.

Solution

(a) Write $p = 4k + 3$ for $k \in \mathbb{Z}_{\geq 0}$. We are given $x^4 \equiv 1 \pmod p$. Note that since x^4 leaves a non-zero residue modulo p, the numbers x and p must be coprime. Thus by Fermat's little theorem

$$\begin{aligned} x^{4k+2} &\equiv 1 \pmod p \\ (x^4)^k \cdot x^2 &\equiv 1 \pmod p \\ 1 \cdot x^2 &\equiv 1 \pmod p \end{aligned}$$

and so $x^2 \equiv 1 \pmod p$ as desired. □

(b) *Solution 1.* Suppose for the sake of contradiction that $x^2 \equiv -1 \pmod p$. We find that

$$\begin{aligned} x^2 \cdot x^2 &\equiv (-1) \cdot (-1) \pmod p \\ x^4 &\equiv 1 \pmod p \end{aligned}$$

However in part (a) we showed that if $x^4 \equiv 1 \pmod p$ then $x^2 \equiv 1 \pmod p$, and since $1 \equiv -1 \pmod p$ only when $p = 2$ we have a contradiction. □

Solution 2. (In this solution do not interpret (a/b) as a fraction. It is instead the Legendre symbol.)

It suffices to show that -1 cannot be a quadratic residue mod p. Let a be a positive integer and p an odd prime. Suppose that a and p are coprime (I'm purposefully avoiding the $a \equiv 0 \bmod p$ case). The Legendre symbol is defined like so

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } a \text{ is not a quadratic residue mod } p \end{cases}$$

Now let $p = 4k + 3$ for some $k \in \mathbb{Z}_{\geq 0}$. By Euler's criterion we note that

$$\begin{aligned} \left(\frac{p-1}{p}\right) &\equiv (-1)^{\frac{p-1}{2}} \pmod p \\ &\equiv (p-1)^{2k+1} \pmod p \\ &\equiv -1 \pmod p. \end{aligned}$$

But if $((p-1)/p)$ were equal to 1 then it wouldn't leave a residue of -1 mod p since $1 \equiv -1 \pmod p$ only when $p = 2$. Thus it must be the case that $((p-1)/p) = -1$ and so by definition -1 isn't a quadratic residue mod p. □

(c) Let us suppose for the sake of contradiction that $p_1, p_2, \ldots, p_n$ are all of the primes that are $\equiv 1 \pmod 4$. Consider the number

$$P = (2p_1p_2 \cdots p_n)^2 + 1.$$

Note that $\gcd(P, p_i) = 1$ due to the Euclidean algorithm. Due to part (b), P is never divisible by a prime of the form $4k + 3$. But since P is odd it must then have some odd prime factors of the form $4k + 1$ but none of these prime come from the collection $p_1, \ldots, p_n$, a contradiction. ■